

Caroline Olamojiba Afolabi

Professional summary

Public health and epidemiology researcher with a First-Class B.Sc. (Hons.) in Microbiology and current M.Sc. training in Epidemiology at the University of Port Harcourt. Research experience spans infectious diseases, antimicrobial resistance, vaccination, disease surveillance, systematic reviews, meta-analysis, and public health data analysis. Author/co-author of peer-reviewed publications and systematic reviews, with practical experience supporting public health emergency response and WHO-led vaccination activities in Nigeria. Research interests include infectious disease epidemiology, outbreak investigation, epidemiological modelling, and evidence-based public health interventions.

Research Interests

Infectious Disease Epidemiology; Disease Surveillance; Outbreak Investigation; Epidemiological Modelling; Global Health; Vaccination and Immunization; Antimicrobial Resistance; Public Health Data Science; Systematic Reviews and Meta-analysis.

Education

- 2026 – May 2027 **Master of Science (M.Sc.) Epidemiology**, *University of Port Harcourt*, Rivers State, Nigeria
Current Grade: **72.76% (A)**
- 2019–2024 **B.Sc. (Hons.) Microbiology – First Class**, *Federal University of Agriculture, Abeokuta*, Ogun State, Nigeria
Supervisor: Professor Flora Oluwafemi
Thesis: Microbial quality of herbs
- 2018 – 2019 **A-Level in Microbiology**, *Cambridge Advance Level program-University of Lagos Akoka*, Lagos State, Nigeria
Grade point: **7/15**

Honors & Awards

- 2025 **Senate Annual Prize for First-Class Graduating Student**, Federal University of Agriculture, Abeokuta, Nigeria.

Research Experience

- 2022 – 2023 **Student Industrial Training**, *Veterinary Teaching Hospital, FUNAAB*
- Assisted final year undergraduate student and master's student in microbiology research projects at College of Veterinary Microbiology Laboratory, FUNNAB, Ogun state, Nigeria
 - Contributed to the evaluation of the antimicrobial activity of crude extract of *vernonia amydalina* against salmonella isolated from household cockroaches.
 - Participated in the antibacterial susceptibility testing and isolation of *s.aures* from a pus sample of a West African dwarf sheep suffering from mastitis.
 - Proficient in performing PCV (packed cell volume) tests, enhancing research capabilities, and contributing to comprehensive data collection and analysis.
 - Conducted laboratory procedures including biochemical assay, and culturing under the supervision of senior researchers.

- 2025 **Medical Laboratory Assistant**, *NYSC Orientation Camp – Batch A, Stream 2, Ebonyi State, Nigeria*
- Part of the camp medical team to provide diagnostic support for corps members requiring medical attention.
 - Assisted the laboratory scientist in conducting a range of point-of-care rapid diagnostic tests, including: Malaria RDT, HIV, Hepatitis B & C, and Syphilis tests, Urinalysis, Pregnancy tests, Random and fasting blood glucose tests
 - Participate in sample collection, test preparation, and result documentation.
 - Support infection prevention and control (IPC) measures and daily lab maintenance to ensure a clean and safe diagnostic environment.

Work Experience

- 2025 - 2026 **NYSC Placement**, *Public Health Emergency Operation Center (PHEOC), Osun State Ministry of Health, Osun State, Nigeria*
- Assisted coordination of public health emergency response activities at the state level
 - Assisted in disease surveillance, data collection, analysis, and reporting for priority diseases
 - Participated in outbreak investigation and response, including field supervision and monitoring
 - Collaborated with stakeholders, including the World Health Organization (WHO) and local health authorities, to strengthen public health interventions
 - Prepared situational reports, presentations, and technical documentation for decision-making.
- Jan 2026 - Mar 2026 **Field Volunteer- Measles-Rubella Vaccination Campaign**, *World Health Organization, Osun State, Nigeria*
- Supported implementation of the Measles-Rubella Vaccination Campaign at the community level of Ede North in Osun State
 - Conducted community mobilization and sensitization to improve awareness and vaccine uptake
 - Collected, recorded, and reported vaccination data for daily review and decision-making
 - Identified and reported field challenges, contributing to timely problem-solving during the three-month campaign.

Teaching Experience

- 2017 – 2019 **Class Teacher**, *Kid's For Christ Primary and Secondary School*
- Served as a class teacher, supporting student academic progress and classroom management.
 - Taught Mathematics in junior school and Biology in senior secondary school.
 - Mentored students in preparation for internal and external examinations including BECE, WAEC, and GCE.
 - Administered and designed assignments, homework, and assessments, ensuring fair allocation and grading.
 - Contributed to the implementation of school policies and regulations.
 - Supported student conduct, discipline, and overall well-being.

Professional Training & Certifications

- 2026 **NEBOSH International General Certificate in Occupational Health and Safety (IGC 1 & 2)**.
- 2026 **Health Safety and Environment 1&2**. *Obafemi Awolowo University, Consultancy service center [UNIFECS], Ile-Ife, Osun State, Nigeria.*
- 2024 **Genomic Training on Basic Bioinformatics For Beginners**. *Helix Biogen Institute, International fellow, Indian Council of Medical Research/African Union-STRC.*
- 2024 **Bioinformatics Course 201**. *Genomac institute Inc. USA Incorporated.*

Publications

Peer-Reviewed Publications

- 2024 **Afolabi, C. O.**, & Ogu, C. K. (2024). Evaluation of antimicrobial susceptibility of *Salmonella* isolated from household cockroaches using *Carica papaya* leaf extract. *International Journal of Research and Scientific Innovation*, XI, 1077–1087. doi:10.51244/IJRSl.2024.11110085
- 2026 **Afolabi, C. O.**, Adekunle, J. D., Oyeniran, M. I., Oyelakin, S. O., Ogu, C. K., Ayanlowo, E. J., Robert, C. O., Sule, H. S., Ideh, G. E., Alagbe, S. A., Fagbemiro, O. O., Adeniyi, Y. A., Adegboyega, T. I., Samsudeen, O. O., Badru, K. O., Shakioye, K. O., Alimi, A. T., Amos, A. A., & Ebonyem, B. N. (2026). COVID-19 origins: A mixed-methods meta-analysis of scientific consensus and political polarization narratives. *Open COVID Journal*. <https://doi.org/10.2174/0126669587424966260109112242>.
- 2026 Olasunkanmi, B., Adekunle, J., Oyelakin, S. O., Obaude, A. M., **Afolabi, C. O.**, Oyeniran, M. I., . . . , & Anifowoshe, T. J. (2026). Classification of severe weather conditions in Nigeria: An integrated weather database with machine learning approach. *Engineering, Mathematics and Computer Science Journal (EMACS)*, 8(1), 37–47. <https://doi.org/10.21512/emacsjournal.v8i1.14302>.

Accepted / Forthcoming Publications

- 2026 Toyon, T., **Afolabi, C.**, Anaemene, D. I., & Adekunle, J. (2026). Maternal dietary quality during pregnancy and risk of neurodevelopmental disorders in offspring: A systematic review and meta-analysis. *Preprints*. <https://doi.org/10.20944/preprints202604.0924.v2>. Accepted for publication in the *British Journal of Nutrition*.

Manuscripts Under Review

- 2026 Oyelakin, S., Adekunle, J., Oyeniran, M., Ayanlowo, J., **Afolabi, C. O.**, & Toyon, T. (2026). How social media influences knowledge, attitudes, and behaviour toward environmental pollution among students. *Preprints*. <https://doi.org/10.20944/preprints202601.1947.v1>.
- 2026 Green, J. W., Adekunle, J. D., & **Afolabi, C. O.** (2026). Symptom expression and emotional distress in online mental health narratives. *Preprints*. <https://doi.org/10.20944/preprints202605.0281.v1>.
- 2025 **Afolabi, C. O.**, Adekunle, J. D., Ogu, C. K., Adegboyega, T. I., Amos, A. A., & Ojewale, I. F. (2025). Microbial quality of herbal drink. *BMJ Microbiology*.

Manuscripts in Preparation

- 2026 **Afolabi, C.O.** & Adekunle, J.D. 2025. A study of the awareness rate of breast cancer and breast self-examination among female students in the Federal University of Agriculture.

Research Methods and Technical Skills

Epidemiology	Epidemiological Data Collection and Management, Public Health Surveillance, Outbreak Investigation and Response, Quantitative Research Methods, Scientific Writing and Literature Review, Questionnaire Design, Statistical Analysis, Data Cleaning and Validation, Research Reporting, Systematic Review and Evidence Synthesis, Laboratory Diagnostics and Microbiological Techniques
Statistics/Data Analysis	SPSS, R, Python, Microsoft Excel; data cleaning, descriptive statistics, inferential statistics, regression analysis, data visualization.
Laboratory	Microbiological culture, staining, microscopy, biochemical testing, antimicrobial susceptibility testing, sample processing.
Bioinformatics	Basic sequence analysis, genomic data analysis, BLAST, NCBI resources, MEGA and related bioinformatics tools.

Languages

English (professional proficiency), Yoruba (native)